

lab2-dl

April 18, 2025

1 Lab 2: Image Classification using Feature Extraction and Machine Learning

Objective : The aim of this lab assignment is to classify images of dogs and cats using feature extraction techniques and machine learning classifiers. You will download the dataset, extract features using HOG and SURF, apply classification models, and compare their performances.

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1.1 Necessary Imports

```
[5]: # Install opencv-contrib-python if not already installed (uncomment if needed)
# !pip install opencv-contrib-python

import os
import random
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from sklearn.cluster import MiniBatchKMeans
from sklearn.linear_model import SGDClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tqdm import tqdm

# Set a random seed for reproducibility
random.seed(42)
np.random.seed(42)
```

1.2 Define Image Loading and Preprocessing Functions

Resize images to a fixed size (e.g. 128×128) so that the HOG (and even SIFT extraction) is consistent.

```
[7]: IMG_SIZE = (128, 128) # width x height

def load_and_preprocess_image(path, img_size=IMG_SIZE):
```

```

# Read in BGR (OpenCV default) and convert to grayscale
img = cv2.imread(path)
if img is None:
    return None
img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
# Resize image
img = cv2.resize(img, img_size)
return img

```

1.3 Select 200 Cat and 200 Dog Images

```

[9]: # Define the dataset directory
data_dir = '/kaggle/input/microsoft-catsvsdogs-dataset/PetImages'

# Initialize lists to store file paths
cat_files = []
dog_files = []

# Iterate through the subdirectories (Cat and Dog)
for subdir in os.listdir(data_dir):
    subdir_path = os.path.join(data_dir, subdir)
    if os.path.isdir(subdir_path): # Ensure it's a directory
        for filename in os.listdir(subdir_path):
            if filename.lower().endswith(('.png', '.jpg', '.jpeg')): #check for
↪image files
                filepath = os.path.join(subdir_path, filename)
                if subdir.lower() == 'cat':
                    cat_files.append(filepath)
                elif subdir.lower() == 'dog':
                    dog_files.append(filepath)

# Randomly sample 200 images from each category
sample_size = 200
cat_files = random.sample(cat_files, min(sample_size, len(cat_files))) #sample
↪only if available
dog_files = random.sample(dog_files, min(sample_size, len(dog_files))) #sample
↪only if available

print(f"Number of cat images: {len(cat_files)}")
print(f"Number of dog images: {len(dog_files)}")

```

Number of cat images: 200
 Number of dog images: 200

1.4 Load the images and create labels

- Cats - 0

- Dogs - 1

```
[10]: images = []
labels = []

# Load cat images (label 0)
for f in tqdm(cat_files, desc="Loading cat images"):
    img = load_and_preprocess_image(os.path.join(data_dir, f))
    if img is not None:
        images.append(img)
        labels.append(0)

# Load dog images (label 1)
for f in tqdm(dog_files, desc="Loading dog images"):
    img = load_and_preprocess_image(os.path.join(data_dir, f))
    if img is not None:
        images.append(img)
        labels.append(1)

images = np.array(images)
labels = np.array(labels)
print(f"Total images loaded: {len(images)}")
```

```
Loading cat images: 100%|      | 200/200 [00:01<00:00, 121.24it/s]
Loading dog images: 100%|      | 200/200 [00:01<00:00, 122.89it/s]

Total images loaded: 399
```

seems like 1 image selected is corrupted

1.5 Split into Training and Testing Sets

80/20 split

```
[11]: X_train, X_test, y_train, y_test = train_test_split(
    images, labels, test_size=0.20, random_state=42, stratify=labels
)

print(f"Training set: {X_train.shape[0]} images")
print(f"Testing set: {X_test.shape[0]} images")
```

```
Training set: 319 images
Testing set: 80 images
```

1.6 Feature Extraction

extract features in two ways: HOG and SIFT (using a bag-of-words approach)

1.6.1 HOG Feature Extraction

```
[12]: def extract_hog_features(image):
    # You can adjust the parameters as needed.
    features, hog_image = hog(
        image,
        orientations=9,
        pixels_per_cell=(8, 8),
        cells_per_block=(2, 2),
        block_norm='L2-Hys',
        visualize=True,
        feature_vector=True
    )
    return features

# Extract HOG features for training and testing sets
hog_features_train = [extract_hog_features(img) for img in tqdm(X_train,
    ↪desc="Extracting HOG (train)")]
hog_features_test = [extract_hog_features(img) for img in tqdm(X_test,
    ↪desc="Extracting HOG (test)")]

hog_features_train = np.array(hog_features_train)
hog_features_test = np.array(hog_features_test)

print("HOG feature shape (train):", hog_features_train.shape)
```

```
Extracting HOG (train): 100%|      | 319/319 [00:22<00:00, 14.45it/s]
```

```
Extracting HOG (test): 100%|      | 80/80 [00:05<00:00, 14.23it/s]
```

```
HOG feature shape (train): (319, 8100)
```

1.6.2 SIFT Feature Extraction via Bag-of-Words

Because SIFT produces a variable number of descriptors per image, we will:

1. Extract SIFT descriptors for each image.
2. Cluster all descriptors from the training set using KMeans (to form the “visual vocabulary”).
3. For each image, build a histogram of the visual word occurrences.

Extract SIFT Descriptors for a Single Image

```
[13]: # Create SIFT extractor (requires opencv-contrib-python)
sift = cv2.SIFT_create()

def extract_sift_descriptors(image):
    # image should be grayscale
    keypoints, descriptors = sift.detectAndCompute(image, None)
    return descriptors
```

Collect SIFT Descriptors from All Training Images

```
[14]: all_descriptors = []

# We will also store each image's descriptors in a list for later histogram
# computation.
sift_descriptors_train = []

for img in tqdm(X_train, desc="Extracting SIFT descriptors (train)":
    des = extract_sift_descriptors(img)
    if des is not None:
        sift_descriptors_train.append(des)
        all_descriptors.append(des)
    else:
        sift_descriptors_train.append(np.array([])) # empty array if no
# descriptors

# Stack all descriptors vertically
all_descriptors = np.vstack([des for des in all_descriptors if des is not None
# and len(des) > 0])
print("Total SIFT descriptors in training set:", all_descriptors.shape)
```

```
Extracting SIFT descriptors (train): 100% | 319/319 [00:02<00:00,
130.50it/s]
```

```
Total SIFT descriptors in training set: (41296, 128)
```

Build the Visual Vocabulary (Codebook) Using KMeans clusters = 50

```
[15]: num_clusters = 50
kmeans = MiniBatchKMeans(n_clusters=num_clusters, random_state=42,
# batch_size=1000)
kmeans.fit(all_descriptors)
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/cluster/_kmeans.py:870:
FutureWarning: The default value of `n_init` will change from 3 to 'auto' in
1.4. Set the value of `n_init` explicitly to suppress the warning
warnings.warn(
```

```
[15]: MiniBatchKMeans(batch_size=1000, n_clusters=50, random_state=42)
```

Create a Function to Compute the SIFT Histogram for an Image

```
[16]: def compute_bow_histogram(descriptors, kmeans_model, num_clusters=num_clusters):
    # If no descriptors are found, return a zero histogram
    if descriptors is None or len(descriptors) == 0:
        return np.zeros(num_clusters)
    # Predict the closest cluster for each descriptor
    cluster_assignments = kmeans_model.predict(descriptors)
```

```

hist, _ = np.histogram(cluster_assignments, bins=np.arange(num_clusters+1))
# Optionally normalize the histogram
hist = hist.astype("float")
if hist.sum() != 0:
    hist /= hist.sum()
return hist

```

Compute Bag-of-Words Histograms for Training and Testing Sets

```

[17]: # For training set
sift_features_train = []
for img, descriptors in zip(X_train, sift_descriptors_train):
    hist = compute_bow_histogram(descriptors, kmeans)
    sift_features_train.append(hist)
sift_features_train = np.array(sift_features_train)
print("SIFT (BoW) feature shape (train):", sift_features_train.shape)

```

SIFT (BoW) feature shape (train): (319, 50)

```

[18]: # For the test set, we need to extract SIFT descriptors first:
sift_features_test = []
for img in tqdm(X_test, desc="Extracting SIFT descriptors (test)":
    des = extract_sift_descriptors(img)
    hist = compute_bow_histogram(des, kmeans)
    sift_features_test.append(hist)
sift_features_test = np.array(sift_features_test)
print("SIFT (BoW) feature shape (test):", sift_features_test.shape)

```

Extracting SIFT descriptors (test): 100% | 80/80 [00:00<00:00, 88.94it/s]

SIFT (BoW) feature shape (test): (80, 50)

1.7 Classification with Epoch Tracking

Train two classifiers on each feature set. Because we want to plot accuracy vs. epoch, we use `SGDClassifier` with `partial_fit` which allows us to simulate training over several epochs.

Note: - For Logistic Regression, we use `loss='log'`. - For SVM, we use `loss='hinge'`.

1.7.1 Define a Function for Epoch-wise Training and Evaluation

```

[19]: def train_with_epochs(X_train, y_train, X_test, y_test, loss, epochs=10):
    """
    Train an SGDClassifier with the specified loss over a given number of
    epochs.
    Returns the final classifier and a list of test accuracies at the end of
    each epoch.
    """

```

```

# Standardize features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Initialize classifier. We set warm_start to True (or use partial_fit) to
↪update the model.
clf = SGDClassifier(loss=loss, max_iter=1, tol=None, random_state=42,
↪learning_rate='optimal', shuffle=True)

# For partial_fit we need to provide the list of classes.
classes = np.unique(y_train)

test_accuracies = []
for epoch in range(epochs):
    # Use partial_fit to train on the whole training set once.
    clf.partial_fit(X_train_scaled, y_train, classes=classes)
    test_accuracy = clf.score(X_test_scaled, y_test)
    test_accuracies.append(test_accuracy)
    print(f"Epoch {epoch+1}/{epochs} - Test Accuracy: {test_accuracy:.4f}")
return clf, test_accuracies, scaler

```

1.7.2 Train on HOG Features

HOG + Logistic Regression

```

[20]: print("Training HOG + Logistic Regression")
hog_lr_clf, hog_lr_accuracies, hog_scaler = train_with_epochs(
    hog_features_train, y_train, hog_features_test, y_test, loss='log',
    ↪epochs=10
)

```

```

Training HOG + Logistic Regression
Epoch 1/10 - Test Accuracy: 0.6875
Epoch 2/10 - Test Accuracy: 0.6875
Epoch 3/10 - Test Accuracy: 0.6250
Epoch 4/10 - Test Accuracy: 0.6625
Epoch 5/10 - Test Accuracy: 0.6375
Epoch 6/10 - Test Accuracy: 0.6375
Epoch 7/10 - Test Accuracy: 0.6375
Epoch 8/10 - Test Accuracy: 0.6375
Epoch 9/10 - Test Accuracy: 0.6375
Epoch 10/10 - Test Accuracy: 0.6375

```

```

/usr/local/lib/python3.10/dist-
packages/sklearn/linear_model/_stochastic_gradient.py:163: FutureWarning: The
loss 'log' was deprecated in v1.1 and will be removed in version 1.3. Use
`loss='log_loss'` which is equivalent.
warnings.warn(

```

HOG + SVM

```
[21]: print("Training HOG + SVM")
hog_svm_clf, hog_svm_accuracies, _ = train_with_epochs(
    hog_features_train, y_train, hog_features_test, y_test, loss='hinge',
    epochs=10
)
```

Training HOG + SVM

```
Epoch 1/10 - Test Accuracy: 0.6375
Epoch 2/10 - Test Accuracy: 0.6375
Epoch 3/10 - Test Accuracy: 0.6750
Epoch 4/10 - Test Accuracy: 0.6750
Epoch 5/10 - Test Accuracy: 0.7000
Epoch 6/10 - Test Accuracy: 0.7000
Epoch 7/10 - Test Accuracy: 0.7125
Epoch 8/10 - Test Accuracy: 0.7000
Epoch 9/10 - Test Accuracy: 0.7000
Epoch 10/10 - Test Accuracy: 0.7000
```

1.7.3 Train on SIFT (Bag-of-Words) Features

SIFT + Logistic Regression

```
[22]: print("Training SIFT + Logistic Regression")
sift_lr_clf, sift_lr_accuracies, sift_scaler = train_with_epochs(
    sift_features_train, y_train, sift_features_test, y_test, loss='log',
    epochs=10
)
```

Training SIFT + Logistic Regression

```
Epoch 1/10 - Test Accuracy: 0.4875
Epoch 2/10 - Test Accuracy: 0.5625
Epoch 3/10 - Test Accuracy: 0.5625
Epoch 4/10 - Test Accuracy: 0.5875
Epoch 5/10 - Test Accuracy: 0.4750
Epoch 6/10 - Test Accuracy: 0.5250
Epoch 7/10 - Test Accuracy: 0.4750
Epoch 8/10 - Test Accuracy: 0.5000
Epoch 9/10 - Test Accuracy: 0.4500
Epoch 10/10 - Test Accuracy: 0.5000
```

/usr/local/lib/python3.10/dist-

```
packages/sklearn/linear_model/_stochastic_gradient.py:163: FutureWarning: The
loss 'log' was deprecated in v1.1 and will be removed in version 1.3. Use
`loss='log_loss'` which is equivalent.
  warnings.warn(
```

SIFT + SVM


```
[23]: print("Training SIFT + SVM")
      sift_svm_clf, sift_svm_accuracies, _ = train_with_epochs(
          sift_features_train, y_train, sift_features_test, y_test, loss='hinge',
          epochs=10
      )
```

```
Training SIFT + SVM
Epoch 1/10 - Test Accuracy: 0.4750
Epoch 2/10 - Test Accuracy: 0.5000
Epoch 3/10 - Test Accuracy: 0.4500
Epoch 4/10 - Test Accuracy: 0.5625
Epoch 5/10 - Test Accuracy: 0.4250
Epoch 6/10 - Test Accuracy: 0.4250
Epoch 7/10 - Test Accuracy: 0.5500
Epoch 8/10 - Test Accuracy: 0.5250
Epoch 9/10 - Test Accuracy: 0.5000
Epoch 10/10 - Test Accuracy: 0.5625
```

1.8 Performance Comparison

```
[24]: print("Final Test Accuracies:")
      print(f"HOG + Logistic Regression: {hog_lr_accuracies[-1]:.4f}")
      print(f"HOG + SVM: {hog_svm_accuracies[-1]:.4f}")
      print(f"SIFT + Logistic Regression: {sift_lr_accuracies[-1]:.4f}")
      print(f"SIFT + SVM: {sift_svm_accuracies[-1]:.4f}")
```

```
Final Test Accuracies:
HOG + Logistic Regression: 0.6375
HOG + SVM: 0.7000
SIFT + Logistic Regression: 0.5000
SIFT + SVM: 0.5625
```

1.9 Feature Fusion & Classification

concatenate the HOG and SIFT feature vectors (for each image) to form a combined feature vector, and then retrain the classifiers.

1.9.1 Create Combined Feature Vectors

```
[25]: # Concatenate features along the feature dimension
      fusion_features_train = np.hstack([hog_features_train, sift_features_train])
      fusion_features_test = np.hstack([hog_features_test, sift_features_test])

      print("Fusion feature shape (train):", fusion_features_train.shape)
      print("Fusion feature shape (test):", fusion_features_test.shape)
```

```
Fusion feature shape (train): (319, 8150)
Fusion feature shape (test): (80, 8150)
```

1.9.2 Train on Fused Features

Fusion + Logistic Regression

```
[26]: print("Training Fusion + Logistic Regression")
      fusion_lr_clf, fusion_lr_accuracies, fusion_scaler = train_with_epochs(
          fusion_features_train, y_train, fusion_features_test, y_test, loss='log',
          ↪epochs=10
      )
```

Training Fusion + Logistic Regression

Epoch 1/10 - Test Accuracy: 0.6250
Epoch 2/10 - Test Accuracy: 0.6500
Epoch 3/10 - Test Accuracy: 0.6875
Epoch 4/10 - Test Accuracy: 0.6875
Epoch 5/10 - Test Accuracy: 0.6875
Epoch 6/10 - Test Accuracy: 0.6875
Epoch 7/10 - Test Accuracy: 0.6750
Epoch 8/10 - Test Accuracy: 0.6750
Epoch 9/10 - Test Accuracy: 0.6750
Epoch 10/10 - Test Accuracy: 0.6750

/usr/local/lib/python3.10/dist-

packages/sklearn/linear_model/_stochastic_gradient.py:163: FutureWarning: The loss 'log' was deprecated in v1.1 and will be removed in version 1.3. Use `loss='log\_loss'` which is equivalent.

warnings.warn(

Fusion + SVM

```
[29]: print("Training Fusion + SVM")
      fusion_svm_clf, fusion_svm_accuracies, _ = train_with_epochs(
          fusion_features_train, y_train, fusion_features_test, y_test, loss='hinge',
          ↪epochs=10
      )
```

Training Fusion + SVM

Epoch 1/10 - Test Accuracy: 0.6250
Epoch 2/10 - Test Accuracy: 0.7000
Epoch 3/10 - Test Accuracy: 0.6375
Epoch 4/10 - Test Accuracy: 0.6750
Epoch 5/10 - Test Accuracy: 0.6875
Epoch 6/10 - Test Accuracy: 0.6625
Epoch 7/10 - Test Accuracy: 0.6625
Epoch 8/10 - Test Accuracy: 0.6625
Epoch 9/10 - Test Accuracy: 0.6625
Epoch 10/10 - Test Accuracy: 0.6625

1.9.3 Report Final Fusion Model Accuracies

```
[30]: print("Final Fusion Test Accuracies:")
      print(f"Fusion + Logistic Regression: {fusion_lr_accuracies[-1]:.4f}")
      print(f"Fusion + SVM: {fusion_svm_accuracies[-1]:.4f}")
```

```
Final Fusion Test Accuracies:
Fusion + Logistic Regression: 0.6750
Fusion + SVM: 0.6625
```

1.10 Plot Epoch vs. Accuracy for Each Combination

```
[31]: epochs = np.arange(1, 11)

plt.figure(figsize=(10, 6))
plt.plot(epochs, hog_lr_accuracies, marker='o', label='HOG + Logistic_
↳Regression')
plt.plot(epochs, hog_svm_accuracies, marker='o', label='HOG + SVM')
plt.plot(epochs, sift_lr_accuracies, marker='o', label='SIFT + Logistic_
↳Regression')
plt.plot(epochs, sift_svm_accuracies, marker='o', label='SIFT + SVM')
plt.plot(epochs, fusion_lr_accuracies, marker='o', label='Fusion + Logistic_
↳Regression')
plt.plot(epochs, fusion_svm_accuracies, marker='o', label='Fusion + SVM')
plt.xlabel("Epoch")
plt.ylabel("Test Accuracy")
plt.title("Epoch vs. Test Accuracy for Different Feature-Classfier_
↳Combinations")
plt.legend()
plt.grid(True)
plt.show()
```

